

DURGA PRASAD TEKUMUDI

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Summary

Results-oriented Machine Learning Operations Engineer with 5 years of experience in designing scalable machine learning pipelines, automating and deploying them in production using CI/CD. Proficient in Python, SQL, Version Control for experimenting, integrating, and deploying, monitoring ML models in Production environment. I am skilled in cloud platforms like AWS and Azure for ML model development and deployment. Experienced with data visualization tools for insightful dashboards. Strong understanding of data governance and security best practices. Collaborative approach to meet cross-functional data needs. Keen desire to learn technical aspects while working in favorable environment.

Skills

- **Programming Languages:** Python (5 years of experience), SQL, R, SAS
- **Web Components:** HTML, CSS, JavaScript, Bootstrap, Angular, React
- **Cloud and Big Data Technologies:** AWS Kinesis Firehose, SageMaker, Kafka, Big Query, Kubeflow.
- **Databases:** PostgreSQL, MongoDB, Snowflake, Redshift, Databricks, Cassandra, Vector Databases.
- **Machine Learning:** Regression, Classification, Clustering, Neural Networks, Sentiment Analysis
- **Data Engineering Tools:** Apache NiFi, Talend, Informatica, Hevo Data, Matillion, Pentaho Data Integration
- **Other Skills:** ETL/ELT, Data Warehousing, Kubernetes, Docker, APIs
- **Soft Skills:** Communication, Collaboration, Problem-Solving, Time Management

Experience

ML Engineer - *SiriusXM*

Feb 2024 -Present

- Designed and deployed a NLP model using BERT for classifying customer interactions (Calls, feedback forms, transcribed calls) as complaints or non-complaints. Improving customer experience and resolving customer issues by 25%.
- Established robust MLOps training pipelines using tools such as Kubeflow, MLflow, DVC for automating model training, versioning, continuous delivery and monitoring of complaint decision system.
- Developed and automated feature engineering pipelines from chat history, feedback forms to streamline text-processing, labelling (complaint vs non-complaint).
- Deployed the trained BERT model to production using cloud-based orchestration platforms (Kubernetes, Docker), enabling real-time/batch inference pipeline on incoming customer data, ensuring high availability, fault tolerance, and quick response times.

ML Engineer - *Micronox Technologies, Hyderabad, Telangana*

May 2019 - July 2021

- Designed and deployed an API that leverages large Language Models (LLM) such as Llama3/Claude Haiku models to provide context-aware responses, enabling users to query and retrieve relevant databases tables and column name from over 2300 tables with 10,000+ columns.
- Applied LLMops principles to fine tune and deploy large language models with optimized retrieval capabilities, ensuring high availability and low-latency responses to complex data queries from 15 years' worth of historical data across numerous tables, significantly improving team productivity and decision-making speed.
- Built and maintained a CI/CD pipeline for automating model retraining and evaluation, incorporating new schema changes to improve the accuracy and relevance of the LLMs data retrieval, ensuring up-to-date and reliable context-based query results.
- Designed and developed scalable model training pipelines using **AWS SageMaker, Kubeflow** and stored the resultant model to the model registry using CICD in dev, stage, and production environments.
- Built batch **inference pipeline** for **RAGs** using the managed Apache Airflow, SageMaker AI, **Bedrock (Llama 3 70B/Claude Sonnet/Claude Haiku)**.

Data Analyst - *Connectial Infosolutions Pvt Ltd, Noida*

July 2018 – Apr 2019

- Developed and maintained real-time data pipelines using **Amazon Kinesis Firehose and Kinesis Data Streams** to process **500,000** data points daily from 100+ IoT devices and web applications, enabling continuous data feed for ML model training and predictions.
- Designed a 10TB data lake architecture on **Amazon S3, optimizing data storage and retrieval for ML applications**, improving data accessibility and processing efficiency by 55%, and accelerating ML model iteration cycles.
- Established robust data governance frameworks across AWS environments to secure and ensure compliance for 200+ data sources, **reducing data access issues by 40% and enabling reliable ML dataset management**.
- Monitored and fine-tuned data pipelines using **Amazon CloudWatch, improving processing speed by 30%, reducing pipeline downtime by 25%**, and ensuring seamless delivery of high-quality training datasets for ML models.
- Streamlined the preparation and integration of real-time and batch datasets, ensuring accurate and consistent inputs for ML workflows, boosting model training efficiency and reducing error rates.

Education

Master Of Science –Applied Statistics and Analytics

Central Michigan University - Michigan, USA

Bachelor Of Technology - Mechanical Engineering

JNTU Hyderabad - India